AI-GENERATED ENGINEERING PROTOTYPE — DAB-V1 — HUMAN REVIEW REQUIRED — NOT HARDWARE QUALIFIED

AIPE 10 kW dual active bridge, 800 V primary / 400 V secondary, 100 kHz, single-phase-shift: generation DAB-V1.
Requirement files: examples/dab-10kw/requirements; insulation table values are reference inputs to be reviewed.
No fabrication, energisation or procurement is authorised by this schematic.

12 V auxiliary input protection, 5 V buck, 3.3 V LDO and supervisor

AUX_FLT_N
NRST

文件: 01_supply.kicad_sch

TMS320F280039C controlCARD socket and signal assignment

CAN_RX
FAULT_RD
RS485_RX

ADC_5V
ADC_ITANK_P
ADC_ITANK_S
ADC_TEMP_P
ADC_TEMP_S
ADC_V12
ADC_VBUS_P
ADC_VBUS_S
ARM
BIAS_EN_MCU
CAN_TX
DRV_RST_MCU
EMU0
EMU1
EPWM1A
EPWM1B
EPWM2A
EPWM2B
EPWM3A
EPWM3B
EPWM4A
EPWM4B
GPIO_LED_RUN
RS485_DE
RS485_TX
RUN_EN
SDFM_P_CLK
SDFM_P_D
SDFM_S_CLK
SDFM_S_D
TCK
TDI
TDO
TEMP_PWM_P_AH
TEMP_PWM_P_AL
TEMP_PWM_P_BH
TEMP_PWM_P_BL
TEMP_PWM_S_AH
TEMP_PWM_S_AL
TEMP_PWM_S_BH
TEMP_PWM_S_BL
TMS
TRST_N
TZ_TRIP
XRS_CARD

文件: 02_card.kicad_sch

Hardware PWM gating by the armed latch

PWM_ENABLE

EPWM1A
EPWM1B
EPWM2A
EPWM2B
EPWM3A
EPWM3B
EPWM4A
EPWM4B
P_PWM_AH
P_PWM_AL
P_PWM_BH
P_PWM_BL
S_PWM_AH
S_PWM_AL
S_PWM_BH
S_PWM_BL

文件: 03_pwm_gating.kicad_sch

Hardware fault tree, over-voltage comparators and fault latch

AUX_FLT_N
NRST
P_VBUS_RX
S_VBUS_RX

ARM
FAULT_RD
PWM_ENABLE
P_FLT_AH_N
P_FLT_AL_N
P_FLT_BH_N
P_FLT_BL_N
P_OC_TRIP_N
P_PG_BIAS_N
P_RDY_AH
P_RDY_AL
P_RDY_BH
P_RDY_BL
RUN_EN
S_FLT_AH_N
S_FLT_AL_N
S_FLT_BH_N
S_FLT_BL_N
S_OC_TRIP_N
S_PG_BIAS_N
S_RDY_AH
S_RDY_AL
S_RDY_BH
S_RDY_BL
TRIPPED
TZ_TRIP

文件: 04_fault_logic.kicad_sch

Isolated-voltage receivers, current buffers and ADC conditioning

NRST

ADC_5V
ADC_ITANK_P
ADC_ITANK_S
ADC_TEMP_P
ADC_TEMP_S
ADC_V12
ADC_VBUS_P
ADC_VBUS_S
BIAS_EN
BIAS_EN_MCU
DRV_RST
DRV_RST_MCU
P_ISNS
P_SDFM_CLK
P_SDFM_D
P_TEMP_HS
P_TEMP_PWM_AH
P_TEMP_PWM_AL
P_TEMP_PWM_BH
P_TEMP_PWM_BL
P_VBUS_RX
P_VSNS_N
P_VSNS_P
SDFM_P_CLK
SDFM_P_D
SDFM_S_CLK
SDFM_S_D
S_ISNS
S_SDFM_CLK
S_SDFM_D
S_TEMP_HS
S_TEMP_PWM_AH
S_TEMP_PWM_AL
S_TEMP_PWM_BH
S_TEMP_PWM_BL
S_VBUS_RX
S_VSNS_N
S_VSNS_P
TEMP_PWM_P_AH
TEMP_PWM_P_AL
TEMP_PWM_P_BH
TEMP_PWM_P_BL
TEMP_PWM_S_AH
TEMP_PWM_S_AL
TEMP_PWM_S_BH
TEMP_PWM_S_BL
XRS_CARD

文件: 05_analog.kicad_sch

CAN, optional RS-485 and JTAG

CAN_RX
CAN_TX
EMU0
EMU1
RS485_DE
RS485_RX
RS485_TX
TCK
TDI
TDO
TMS
TRST_N

文件: 06_comms.kicad_sch

Power-board harnesses, indicators, test points and mounting

TRIPPED

BIAS_EN
DRV_RST
GPIO_LED_RUN
P_FLT_AH_N
P_FLT_AL_N
P_FLT_BH_N
P_FLT_BL_N
P_ISNS
P_OC_TRIP_N
P_PG_BIAS_N
P_PWM_AH
P_PWM_AL
P_PWM_BH
P_PWM_BL
P_RDY_AH
P_RDY_AL
P_RDY_BH
P_RDY_BL
P_SDFM_CLK
P_SDFM_D
P_TEMP_HS
P_TEMP_PWM_AH
P_TEMP_PWM_AL
P_TEMP_PWM_BH
P_TEMP_PWM_BL
P_VSNS_N
P_VSNS_P
S_FLT_AH_N
S_FLT_AL_N
S_FLT_BH_N
S_FLT_BL_N
S_ISNS
S_OC_TRIP_N
S_PG_BIAS_N
S_PWM_AH
S_PWM_AL
S_PWM_BH
S_PWM_BL
S_RDY_AH
S_RDY_AL
S_RDY_BH
S_RDY_BL
S_SDFM_CLK
S_SDFM_D
S_TEMP_HS
S_TEMP_PWM_AH
S_TEMP_PWM_AL
S_TEMP_PWM_BH
S_TEMP_PWM_BL
S_VSNS_N
S_VSNS_P

文件: 07_interface.kicad_sch

AI-GENERATED ENGINEERING PROTOTYPE — DAB-V1 — HUMAN REVIEW REQUIRED — NOT HARDWARE QUA

Sheet: /
File: dab-control.kicad_sch

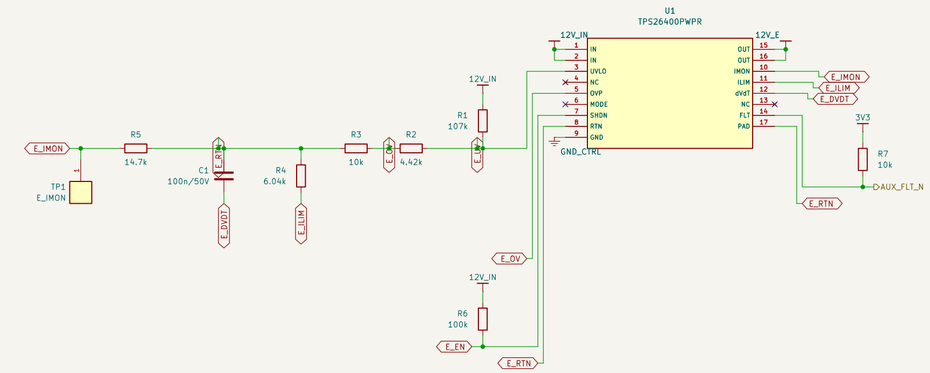
Title: AIPE 10 kW DAB — control board (TMS320F280039C controlCARD)

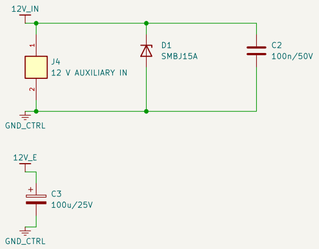
Size: A1 | Date: | Rev: DAB-V1

KiCad E.D.A. 10.0.6 | Id: 1/8

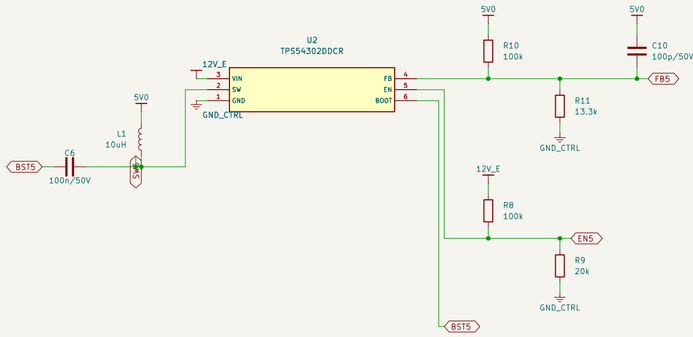
12 V auxiliary input protection, 5 V buck, 3.3 V LDO and supervisor

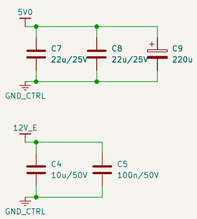
12 V auxiliary entry: TVS and TPS2640 eFuse (CB-PR-001)



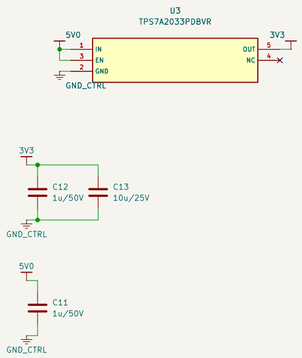


5 V buck for the controlCARD, logic and transceivers

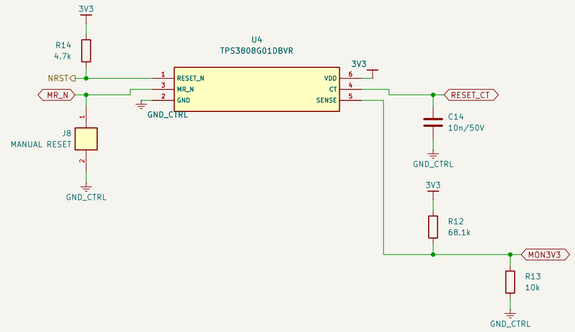




3.3 V LDO for the on-board logic



3.3 V supervisor: reset holds the fault latch cleared





Supply provenance (PWR_FLAG)



AI-GENERATED ENGINEERING PROTOTYPE – DAB-V1 – HUMAN REVIEW REQUIRED – NOT HARDWARE QUA

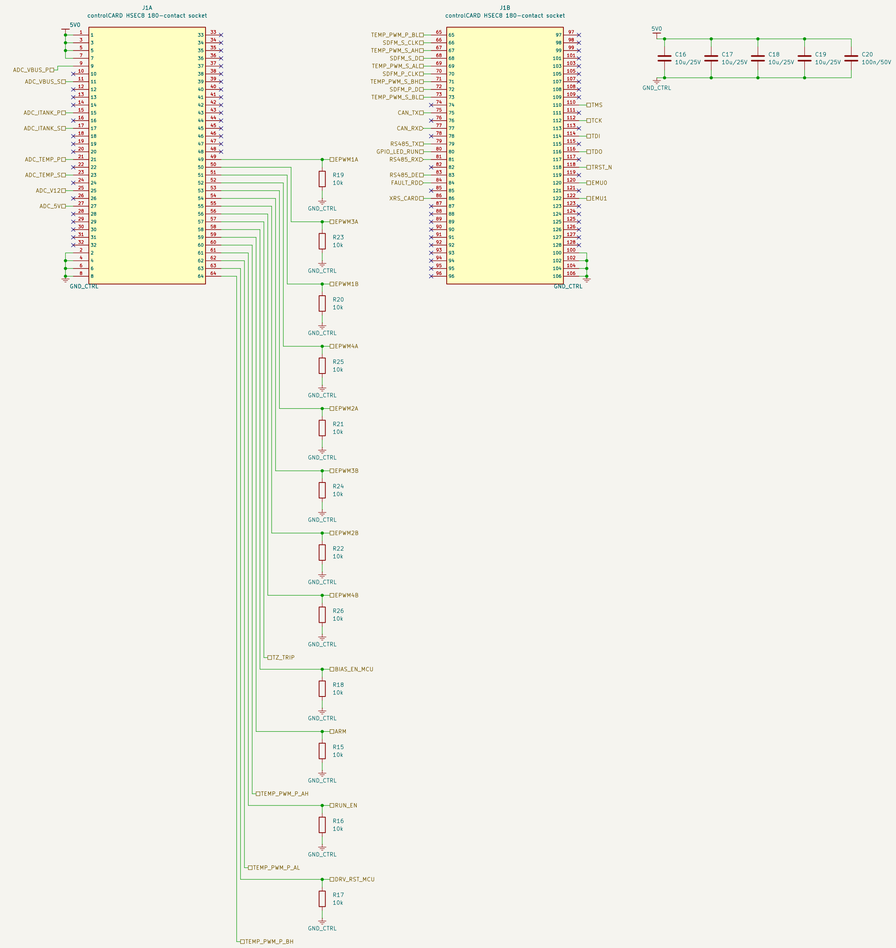
Sheet: /12 V auxiliary input protection, 5 V buck, 3.3 V LDO and supervisor/
File: 01_supply.kicad_sch

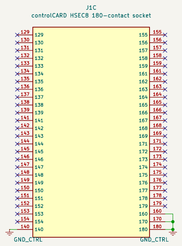
Title: 12 V auxiliary input protection, 5 V buck, 3.3 V LDO and supervisor

Size: A2	Date:	Rev: DAB-V1
KiCad E.D.A. 10.0.6		Id: 2/8

TMS320F280039C controlCARD socket and signal assignment

TMS320F280039C controlCARD socket (signal map ASSUMED — review item R-01)



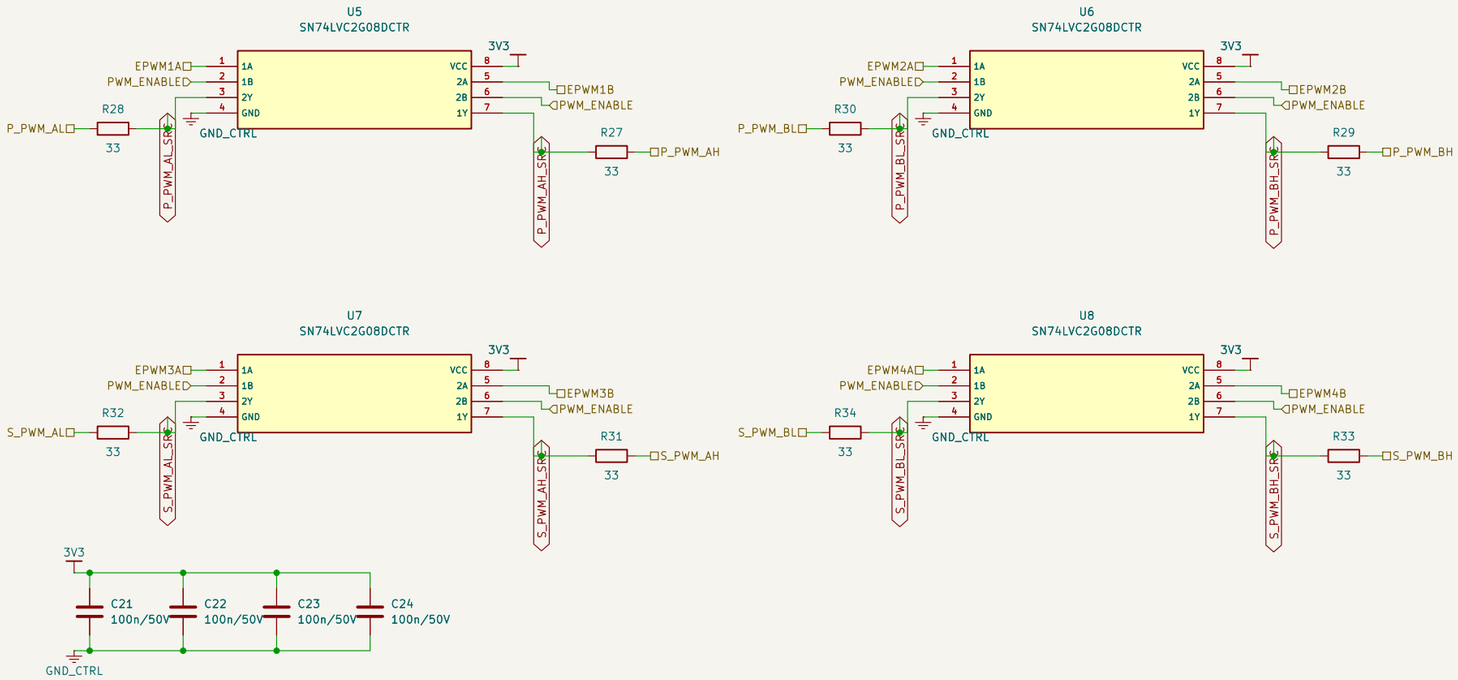


AI-GENERATED ENGINEERING PROTOTYPE — DAB-V1 — HUMAN REVIEW REQUIRED — NOT HARDWARE QUA

Sheet: /TMS320F280039C controlCARD socket and signal assignment/	
File: 02_card.kicad_sch	
Title: TMS320F280039C controlCARD socket and signal assignment	
Size: A1 — Date:	**Rev: DAB-V1**
KiCad E.D.A. 10.0.6	Id: 3/8

Hardware PWM gating by the armed latch

Every PWM is ANDed with the hardware PWM_ENABLE before it leaves the board



AI-GENERATED ENGINEERING PROTOTYPE – DAB-V1 – HUMAN REVIEW REQUIRED – NOT HARDWARE QUA

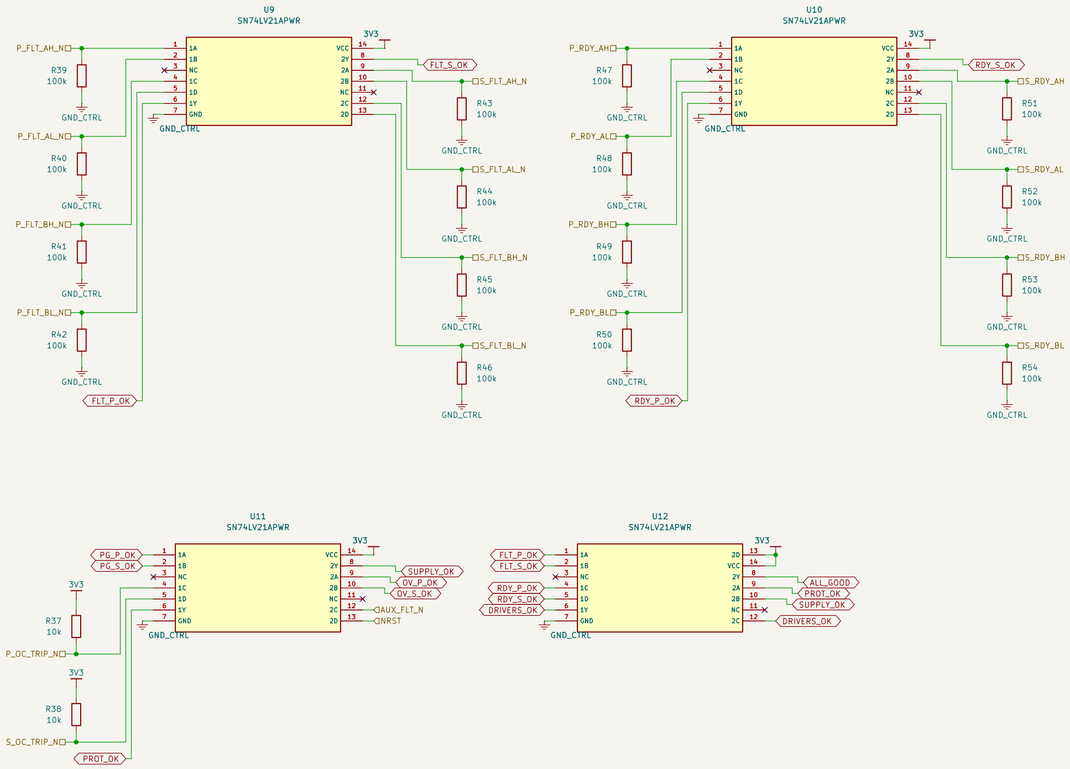
Sheet: /Hardware PWM gating by the armed latch/
File: 03_pwm_gating.kicad_sch

Title: Hardware PWM gating by the armed latch

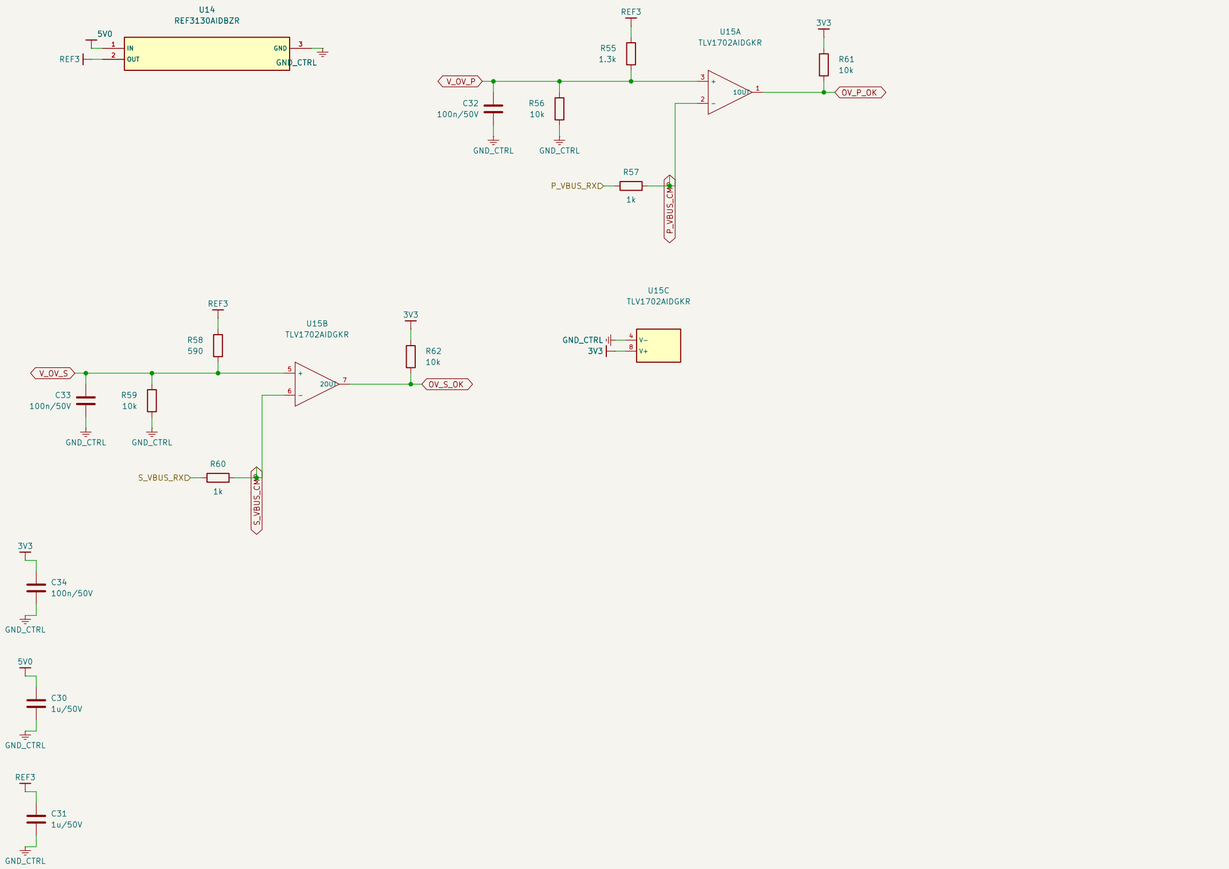
Size: A3	Date:	**Rev: DAB-V1**
KiCad E.D.A. 10.0.6		Id: 4/8

Hardware fault tree, over-voltage comparators and fault latch

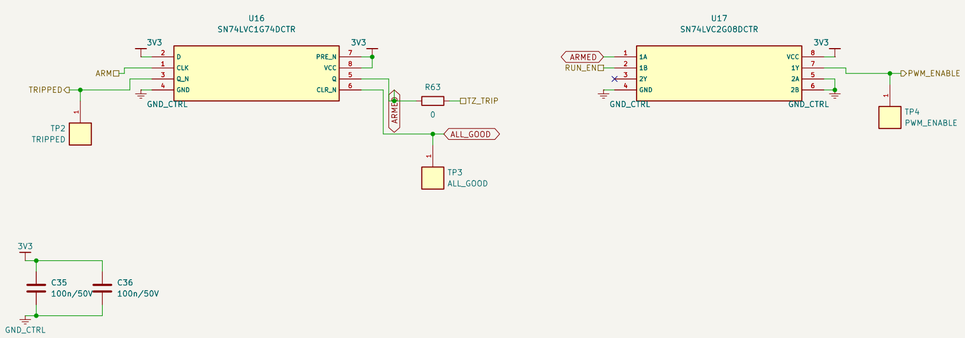
Asynchronous hardware fault tree (KR-PR-002): drivers, bias power-good, over-current, over-voltage, supplies



Bus over-voltage comparators on the received isolated signals



Armed latch: asynchronous clear has priority; re-arm needs an edge after every cause has cleared



AI-GENERATED ENGINEERING PROTOTYPE — DAB-V1 — HUMAN REVIEW REQUIRED — NOT HARDWARE QUA

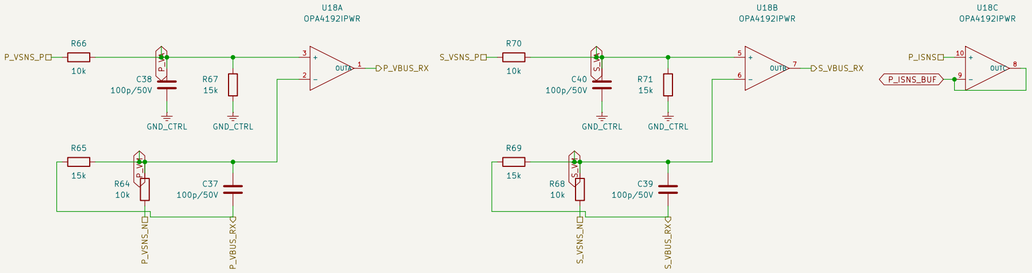
Sheet: /Hardware fault tree, over-voltage comparators and fault latch/
File: 04_fault_logic.kicad_sch

Title: Hardware fault tree, over-voltage comparators and fault latch

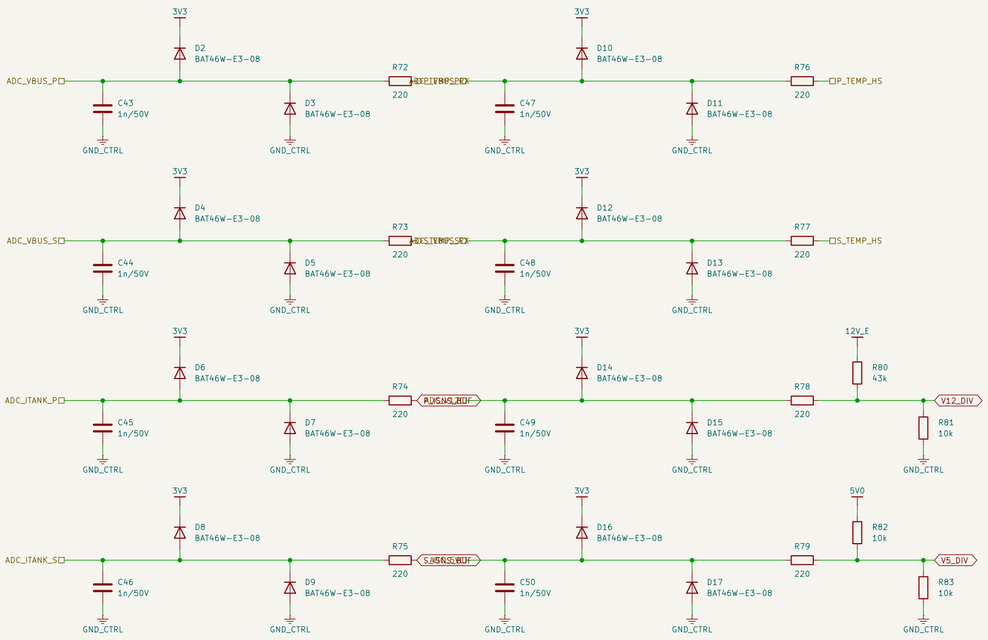
Size: A2	Date:	Rev: DAB-V1
KiCad E.D.A. 10.0.6		Id: 5/8

Isolated–voltage receivers, current buffers and ADC conditioning

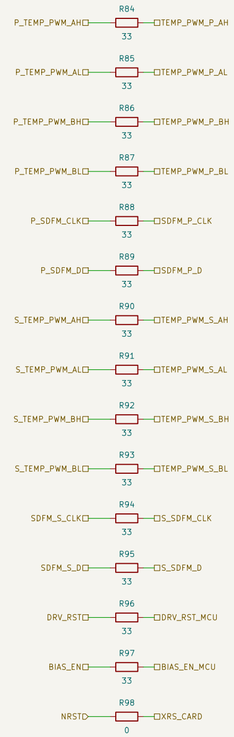
Differential receivers (gain 1.5) for the isolated voltage signals and buffers for the Hall outputs



ADC input RC and Schottky clamps (0.22 us, 3.3 V rails)



Series terminations of the harness digital signals



AI-GENERATED ENGINEERING PROTOTYPE – DAB-V1 – HUMAN REVIEW REQUIRED – NOT HARDWARE QUA

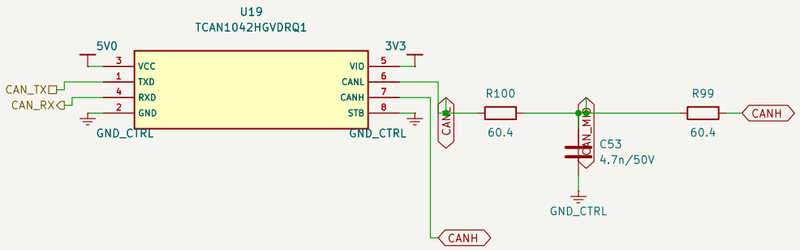
Sheet: /Isolated–voltage receivers, current buffers and ADC conditioning/
File: 05_analog.kicad_sch

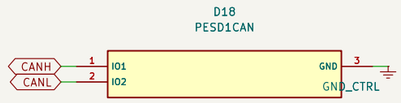
Title: Isolated–voltage receivers, current buffers and ADC conditioning

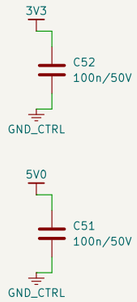
Size: A2	Date:	Rev: DAB-V1
KiCad E.D.A. 10.0.6		Id: 6/8

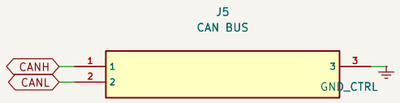
CAN, optional RS-485 and JTAG

CAN FD transceiver with split termination and ESD protection

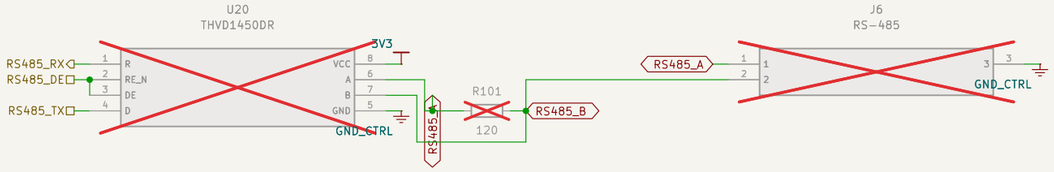




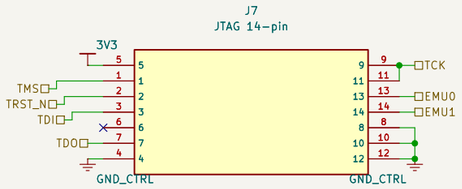




Optional RS-485 half duplex (not populated)



TI 14-pin JTAG / cJTAG header (card debug contacts ASSUMED)



AI-GENERATED ENGINEERING PROTOTYPE – DAB-V1 – HUMAN REVIEW REQUIRED – NOT HARDWARE QUA

Sheet: /CAN, optional RS-485 and JTAG/
File: 06_comms.kicad_sch

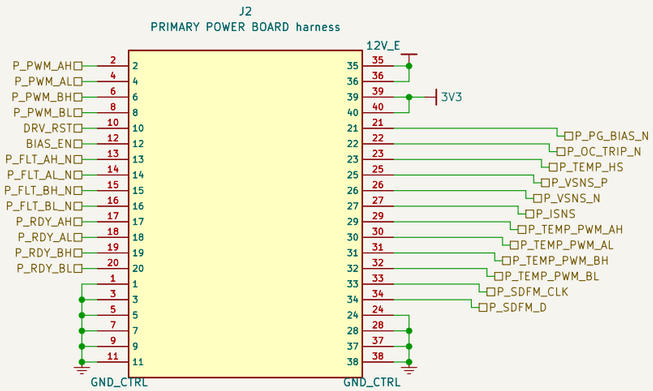
Title: CAN, optional RS-485 and JTAG

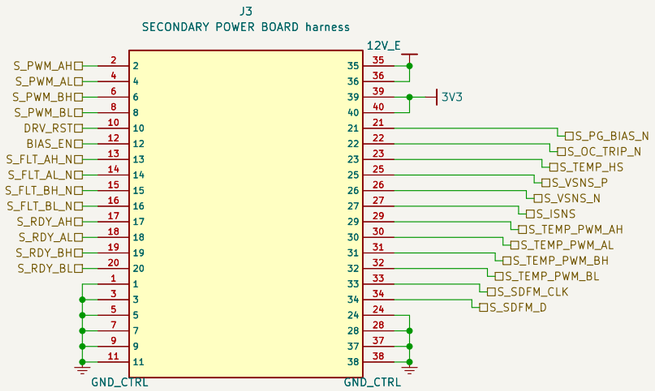
Size: A3	Date:	**Rev: DAB-V1**
KiCad E.D.A. 10.0.6		Id: 7/8

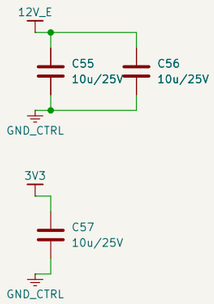
Power-board harnesses, indicators, test points and mounting

PWR_FLAG symbols state supply provenance for ERC; they do not prove voltage, current capacity or protection.

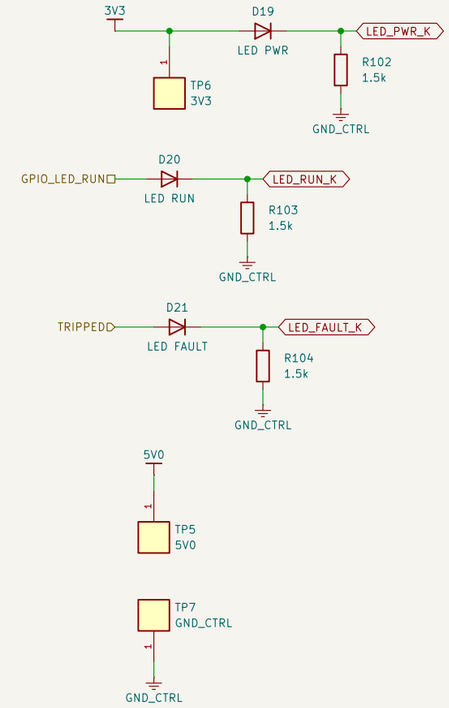
Keyed 40-way harness headers to the primary and secondary power boards



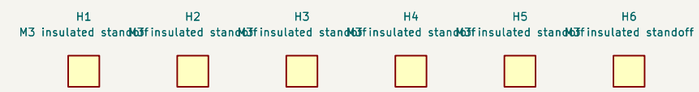




Status indicators



Mechanical



Supply provenance (PWR_FLAG)



AI-GENERATED ENGINEERING PROTOTYPE — DAB-V1 — HUMAN REVIEW REQUIRED — NOT HARDWARE QUA

Sheet: /Power-board harnesses, indicators, test points and mounting/
File: 07_interface.kicad_sch

Title: Power-board harnesses, indicators, test points and mounting

Size: A3	Date:	Rev: DAB-V1
KiCad E.D.A. 10.0.6		Id: 8/8